

2017 NYJC JC2 Preliminary Examination 9758/2 Solution

Qn	Remarks
1(i) Assume that $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors. $\overrightarrow{OC} = 5\mathbf{a}$, $\overrightarrow{OD} = 2\mathbf{b}$ On the line BC, $\overrightarrow{OM} = \lambda(5\mathbf{a}) + (1-\lambda)\mathbf{b}$ On the line AD, $\overrightarrow{OM} = \mu(2\mathbf{b}) + (1-\mu)\mathbf{a}$ Since $\mathbf{a}$ and $\mathbf{b}$ are non-zero, non-parallel vectors, comparing coefficient $5\lambda = 1 - \mu$ $2\mu = 1 - \lambda \Rightarrow \lambda = \frac{1}{9}, \mu = \frac{4}{9}$ Thus $\overrightarrow{OM} = \frac{5}{9}\mathbf{a} + \frac{8}{9}\mathbf{b}$	Note that this is a condition on vectors, not points. You may use Ratio Theorem, or you may find direction vectors $\overrightarrow{BC}$ and $\overrightarrow{AD}$ to find the respective lines.
1(ii) Since $\mathbf{a}$ is a unit vector in the direction of OC, shortest distance $= \left \overrightarrow{OM} \times \mathbf{a} \right$ $= \left \left(\frac{5}{9}\mathbf{a} + \frac{8}{9}\mathbf{b} \right) \times \mathbf{a} \right $ $= \frac{8}{9} \mathbf{a} \times \mathbf{b} $ $k = \frac{8}{9}$	Some students mixed up the properties for cross product and dot product. $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, $\mathbf{a} \cdot \mathbf{a} = \mathbf{a} ^2$ Note that distance cannot be negative. Always check your algebraic workings when your answers appear counter-intuitive.
2(a) For sum to infinity to exist, $\tan \theta < 1$ $-1 < \tan \theta < 1$ $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$ $\frac{1}{1 - \tan \theta} > 2$ $0 < 1 - \tan \theta < \frac{1}{2}$ $\tan \theta > \frac{1}{2} \Rightarrow \theta > 0.464$ Since $-\frac{\pi}{4} < \theta < \frac{\pi}{4}$, therefore $\{\theta \in \mathbb{R} \mid 0.464 < \theta < 0.786\}$ or $\theta : (0.464, 0.786)$	Most students failed to check the range of values for $r < 1$ for sum to infinity to exist. Note: Many students cross multiplied to get $1 > 2(1 - \tan \theta)$. For this case it is ok as $1 - \tan \theta > 0$. In general, we should not cross multiply for inequalities unless the term multiplied is strictly positive. Set notation.

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2(b)	$u_1 = S_1 = 2 \Rightarrow a = 8$ $u_2 = S_2 - S_1 = 10 \Rightarrow d = 8$ $u_{32} = a + (32-1)d = 2 + (32-1)8 = 250$ $\frac{u_{32}}{u_m} = \frac{u_m}{u_2} = \text{constant}$ $\Rightarrow (u_m)^2 = (10)(250) = 2500$ $u_m = 50 \text{ (since it is a positive sequence)}$ $50 = 2 + (m-1)8 \Rightarrow m = 7$ Alternatively, $u_n = S_n - S_{n-1}$ $= 4n^2 - 2n - [4(n-1)^2 - 2(n-1)]$ $= 8n - 6$ $\frac{u_{32}}{u_m} = \frac{u_m}{u_2}$ $\frac{8(32) - 6}{8m - 6} = \frac{8m - 6}{8(2) - 6}$ $(8m - 6)^2 = (250)(10) = 2500$ $m = 7 \text{ or } m = -5.5 \text{ (rejected as } m \text{ is a positive integer)}$	u_2, u_m and u_{32} are consecutive terms of GP
3(i)	$y = \ln(\cos ax - \sin ax)$ $e^y = \cos ax - \sin ax$ $e^y \frac{dy}{dx} = -a \sin ax - a \cos ax$ $e^y \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx} \right)^2 = -a^2 \cos ax + a^2 \sin ax$ $e^y \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx} \right)^2 = -a^2 (\cos ax - \sin ax)$ $e^y \frac{d^2y}{dx^2} + e^y \left(\frac{dy}{dx} \right)^2 = -a^2 e^y$ $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 + a^2 = 0$	A majority of students produced <u>very long, tedious and messy</u> calculations by direct differentiation when the calculation could have been much simpler by rewriting the equation into the implicit form $e^y = \cos ax - \sin ax$ and applying implicit differentiation to obtain the desired equation. Students MUST learn SMART WAYS of doing math instead of using the brute force method

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3(ii)	$\frac{d^3y}{dx^3} + 2\left(\frac{dy}{dx}\right)\frac{d^2y}{dx^2} = 0$ When $x = 0$, $y = 0$ $\frac{dy}{dx} = -a, \quad \frac{d^2y}{dx^2} = -2a^2, \quad \frac{d^3y}{dx^3} = -4a^3$ $y = -ax - a^2x^2 - \frac{2}{3}a^3x^3 + \dots$	Fairly straightforward application of Maclaurin's theorem to obtain series expansion.
3(iii)	$\ln(\cos 2x - \sin 2x) = -2x - 4x^2 - \frac{16}{3}x^3 + \dots$ $\cos 2x - \sin 2x \approx e^{-2x-4x^2}$ $\approx 1 + (-2x - 4x^2) + \frac{(-2x - 4x^2)^2}{2!} \text{ (since } e^x \approx 1 + x + \frac{x^2}{2!} \text{)}$ $\approx 1 - 2x - 4x^2 + \frac{(-2x)^2}{2}$ $= 1 - 2x - 2x^2 \text{ where } k = -2$	A number of students did not pay attention to the word 'Hence' which requires them to use an earlier result to deduce the next result. Many simply used the series expansions of $\sin x$ and $\cos x$ from MF26 which earn no credit
3(iv)	$\cos 2x - \sin 2x = 1 - \frac{(2x)^2}{2} - (2x)$ $= 1 - 2x - 2x^2$	Some students forgot the '2' and wrote $\cos 2x - \sin 2x$ $= 1 - \frac{x^2}{2} - x$ Some used the double angle formulae for $\sin 2x$ and $\cos 2x$ which is not necessary

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4(i)	$\frac{dx}{dt} = k(N-x)x$ $\frac{1}{(N-x)x} \frac{dx}{dt} = k$ $\int \frac{1}{(N-x)x} dx = \int k dt$ $\frac{1}{N} \int \frac{1}{N-x} + \frac{1}{x} dx = \int k dt$ $\frac{1}{N} (-\ln N-x + \ln x) = kt + C$ Remember to include the modulus sign whenever the integral involves ln. $\frac{1}{N} \ln \left \frac{x}{N-x} \right = kt + C$ $\ln \left \frac{x}{N-x} \right = Nkt + NC$ $\left \frac{x}{N-x} \right = e^{Nkt+NC}$ $\frac{x}{N-x} = Ae^{Nkt} \quad \text{where } A = \pm e^{NC}$ When $t=0$, $x = \frac{1}{3}N$, $A = \frac{1}{2}$ $\frac{x}{N-x} = \frac{1}{2} e^{Nkt}$ $2x = (N-x)e^{Nkt}$ $x(2 + e^{Nkt}) = Ne^{Nkt}$ $x = \frac{Ne^{Nkt}}{2 + e^{Nkt}} \quad \text{equivalently, } x = \frac{N}{2e^{-Nkt} + 1}$	
	<u>Alternative method</u> (not recommended):	

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	$\frac{1}{(N-x)x} \frac{dx}{dt} = k$ $-\int \frac{1}{x^2 - Nx} dx = \int k dt$ $-\int \frac{1}{\left(x - \frac{N}{2}\right)^2 - \left(\frac{N}{2}\right)^2} dx = \int k dt$ $-\left(\frac{1}{2\left(\frac{N}{2}\right)} \ln \left \frac{\left(x - \frac{N}{2}\right) - \frac{N}{2}}{\left(x - \frac{N}{2}\right) + \frac{N}{2}} \right \right) = kt + C$ $-\left(\frac{1}{N} \ln \left \frac{x - N}{x} \right \right) = kt + C$ $\frac{1}{N} \ln \left \frac{x}{N - x} \right = kt + C$ $\ln \left(\frac{x}{N - x} \right) = Nkt + NC \quad \text{since } 0 < x < N$ When $t=0$, $x = \frac{1}{3}N$, $\ln \frac{1}{2} = NC \Rightarrow C = -\frac{1}{N} \ln 2$ $\ln \left(\frac{x}{N - x} \right) = Nkt - \ln 2$ $\ln \left(\frac{2x}{N - x} \right) = Nkt$ $\frac{2x}{N - x} = e^{Nkt}$ $x(2 + e^{Nkt}) = N e^{Nkt}$ $x = \frac{N e^{Nkt}}{2 + e^{Nkt}} \quad \text{equivalently, } x = \frac{N}{2e^{-Nkt} + 1}$ As $t \rightarrow \infty$, $x = \frac{N}{2e^{-Nkt} + 1} \rightarrow N$	We can remove the modulus sign by taking into account the range of values the variable x can take

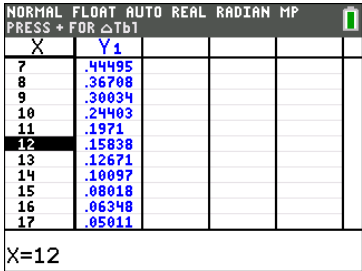
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(ii)	$\frac{d^2 x}{dt^2} = \frac{-9t}{(4+9t^2)^2}$$\int \frac{d^2 x}{dt^2} dt = \int \frac{-9t}{(4+9t^2)^2} dt$$= -\frac{1}{2} \int \frac{18t}{(4+9t^2)^2} dt$$\frac{dx}{dt} = \frac{1}{2} \left(\frac{1}{4+9t^2} \right) + A$$\int \frac{dx}{dt} dt = \frac{1}{2} \int \frac{1}{4+9t^2} dt + \int A dt$$= \frac{1}{18} \int \frac{1}{\left(\frac{2}{3}\right)^2 + t^2} dt + \int A dt$$= \frac{1}{18} \cdot \frac{1}{\frac{2}{3}} \tan^{-1} \left(\frac{t}{\frac{2}{3}} \right) + At + B$$x = \frac{1}{12} \tan^{-1} \left(\frac{3t}{2} \right) + At + B$ Since the population stabilises in the long run, as $t \rightarrow \infty$, $x \rightarrow$ finite value, $A = 0$ When $t = 0$, $x = \frac{1}{3}N$, $B = \frac{N}{3}$ Hence $x = \frac{1}{12} \tan^{-1} \left(\frac{3t}{2} \right) + \frac{N}{3}$ When $t \rightarrow \infty$, $\tan^{-1} \left(\frac{3t}{2} \right) \rightarrow \frac{\pi}{2}$ Hence $x \rightarrow \frac{\pi}{24} + \frac{N}{3}$.	Integral $\int \frac{18t}{(4+9t^2)^2} dt$ is of the form $\int f'(t)[f(t)]^n dt$ where $n = -2$. Some students wrongly express the integral as $\frac{1}{2} \int \frac{1}{4+9t^2} + A dt$. Note that $\int \frac{1}{2} \left(\frac{1}{4+9t^2} \right) + A dt = \frac{1}{2} \int \frac{1}{4+9t^2} + \underline{2}A dt$. Thus it is recommended to express as two separate integrals as shown in the solution. To find $\int \frac{1}{4+9t^2} dt$, it is necessary to have the coefficient of t^2 to be 1 before applying formula in MF26. Thus $\int \frac{1}{4+9t^2} dt = \frac{1}{9} \int \frac{1}{\left(\frac{2}{3}\right)^2 + t^2} dt = \frac{1}{\frac{2}{3}} \tan^{-1} \left(\frac{t}{\frac{2}{3}} \right) = \frac{3}{2} \tan^{-1} \left(\frac{3t}{2} \right)$ and <u>not</u> $\int \frac{1}{4+9t^2} dt = \int \frac{1}{2^2 + (3t)^2} dt = \frac{1}{2} \tan^{-1} \left(\frac{3t}{2} \right)$.												
5(i)	Let Y be the payment for an individual. The probability table is as follows: <table><tr><td>y</td><td>100</td><td>200</td><td>300</td><td>325</td><td>350</td></tr><tr><td>$P(Y = y)$</td><td>$\frac{1}{3}$</td><td>$\frac{4}{15}$</td><td>$\frac{1}{5}$</td><td>$\frac{2}{15}$</td><td>$\frac{1}{15}$</td></tr></table> $E(Y) = 100 \cdot \frac{1}{3} + 200 \cdot \frac{4}{15} + 300 \cdot \frac{1}{5} + 325 \cdot \frac{2}{15} + 350 \cdot \frac{1}{15}$ $= 213\frac{1}{3}$	y	100	200	300	325	350	$P(Y = y)$	$\frac{1}{3}$	$\frac{4}{15}$	$\frac{1}{5}$	$\frac{2}{15}$	$\frac{1}{15}$	Candidates must read the question carefully to understand the payment scale, and thereafter to write down the probability table for Y correctly. Note that there is no linear relationship between Y and X. Thus working out $E(X)$ will not obtain any credit.
y	100	200	300	325	350									
$P(Y = y)$	$\frac{1}{3}$	$\frac{4}{15}$	$\frac{1}{5}$	$\frac{2}{15}$	$\frac{1}{15}$									

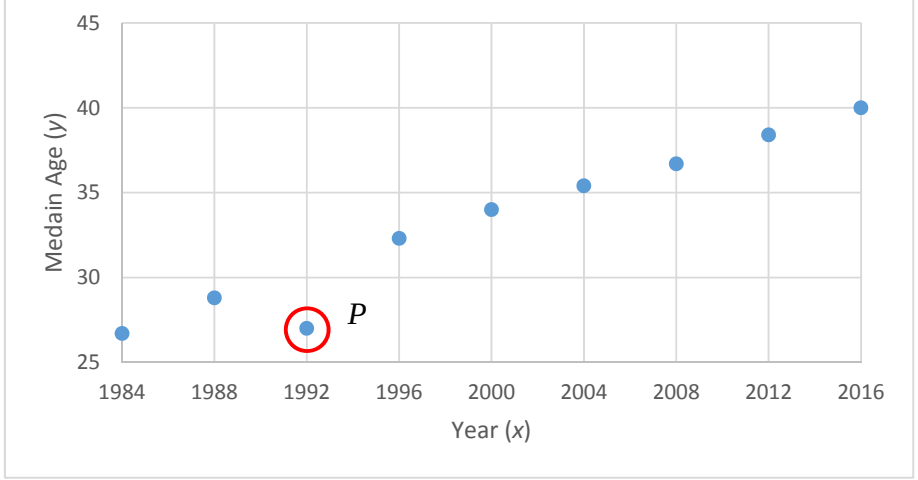
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5(ii)	$E(Y^2) = 100^2 \cdot \frac{1}{3} + 200^2 \cdot \frac{4}{15} + 300^2 \cdot \frac{1}{5} + 325^2 \cdot \frac{2}{15} + 350^2 \cdot \frac{1}{15}$ $= 54250$ $\text{Var}(Y) = E(Y^2) - [E(Y)]^2 = 8738\frac{8}{9}$ Since $n=100$ is large, by Central Limit Theorem, $T = \sum_{i=1}^{100} Y_i \sim N(21333.33, 873888.89) \text{ approx.}$ Prob. Req'd = $P(T > 24000)$ ≈ 0.00217	Note that $E(Y^2)$ is not the variance of Y. Further, candidates are reminded that you should not assume that Y is normally distributed. Central Limit Theorem is necessary to obtain the approximate distribution of ΣY_i.
6	Probability required = $\frac{2 \times {}^3C_2 \times {}^5C_2}{{}^8C_4} = \frac{6}{7}$	${}^3C_2 \times {}^5C_2$ – choose 2 girls from the remaining 3 girls and 2 boys from 5 boys for the group with exactly 3 girls. Multiply by 2 because this group can be Ann or Alice's group. This is a conditional probability as Ann and Alice must be the team leaders, thus 8C_4.
6(i)	Required probability = $\frac{(8-1)! {}^8C_2 2!}{(10-1)!} = \frac{7}{9}$ OR $1 - \frac{(9-1)! 2!}{(10-1)!} = \frac{7}{9}$	Insertion method or apply Principle of complementation

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6(ii)	Let X be the event that the remaining 3 girls are separated. Let Y be the event that Ann and Alice are not seated together. $P(X Y) = \frac{P(X \cap Y)}{P(Y)}$ $= \frac{P(X) - P(X \cap Y')}{P(Y)}$ $= \frac{(7-1)!^7 C_3 3! - (6-1)!2!^6 C_3 3!}{(10-1)!}$ $= \frac{7/9}{7/9}$ $= \frac{85}{196}$	Not possible to list out all the cases.
7(i)	$P(R=6) = 2P(R=4)$ ${}^{10}C_6 p^6 (1-p)^4 = 2 {}^{10}C_4 p^4 (1-p)^6$ $p^2 = 2(1-p)^2$ $p^2 - 4p + 2 = 0$ $p = 0.586$	
7(ii)	$R \sim B(n, 0.25)$ $P(R < 2) > 0.15$ $P(R \leq 1) > 0.15$ $n = 12$ 	

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7(iii)	$R \sim B(11, 0.7)$ $P(R \geq 5 R \leq 8) = \frac{P(R \geq 5 \text{ and } R \leq 8)}{P(R \leq 8)}$ $= \frac{P(5 \leq R \leq 8)}{P(R \leq 8)}$ $= \frac{P(R \leq 8) - P(R \leq 4)}{P(R \leq 8)}$ $= 0.969$	2 major errors seen in the students' scripts. 1. Failure to recognise that this is a conditional probability. 2. Failure to count the number of cases for the numerator.
8(a)(i)	The value of 0.073 indicates that there is a weak linear correlation between petal size and the amount of water but there could be some non-linear relation.	Quite a number of students state that there is still some weak linear correlation.
8(a)(ii)	The approximate value of -1 indicates that there is a strong negative linear correlation between the risk of heart disease and amount of red wine intake. It does not mean that red wine intake decreases the risk of heart disease.	
8(b)(i)		Wrong scale is used by a handful of students.
8(b)(ii)	For Model A, $r = -0.9985438 = 0.99854$ For Model B, $r = 0.9984431 = 0.99844$	Quite a number of students chose model B instead of A simply because r is positive.
8(b)(iii)	Model A as the r value is closer to 1. The suitable regression line is $y = 848.24 - \frac{1629165.57}{x}$	
8(b)(iv)	This is because age (x) is the controlled variable	Badly answered. Students are not able to identify controlled variable.

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8(b) (v)	The rise in the median age is due to the drop in the growth of the population.	
9(i)	To test $H_0 : \mu = 0.5$ $H_1 : \mu \neq 0.5$ Level of significance: 5% Under H_0, $Z = \frac{\bar{X} - 0.5}{0.04 / \sqrt{25}} \sim N(0,1)$ Reject H_0 if p-value ≤ 0.05 Calculation: $\bar{x} = 0.51$, p-value = 0.211 Since p-value > 0.05, we do not reject H_0. Thus there is insufficient evidence at 5% level of significance that the manufacturing process is producing ball bearings of different diameters. Distribution of the diameter of the ball bearings is normal.	On the whole, this question was badly done. Many candidates still showed poor presentation. They have also illustrated poor understanding in the writing of rejection region/criteria. Common mistakes include swapping μ_0 and $\bar{x}$; identify H_1 wrongly.
9(ii)	To test $H_0 : \mu = 0.5$ $H_1 : \mu > 0.5$ Level of significance: 5% Under H_0, by Central Limit Theorem, $Z = \frac{\bar{X} - 0.5}{s / \sqrt{100}} \sim N(0,1)$ approx Reject H_0 if p-value ≤ 0.05 Calculation: $\bar{x} = 0.506$, $\Sigma(x - 0.5) = 50.6 - 50 = 0.6$ $s^2 = \frac{1}{n-1} \left[\Sigma(x-0.5)^2 - \frac{(\Sigma(x-0.5))^2}{n} \right]$ $= \frac{1}{99} \left(0.08345 - \frac{0.6^2}{100} \right)$ $= 8.07 \times 10^{-4}$ p-value = 0.0173 Since p-value < 0.05, we reject H_0. Thus there is sufficient evidence at 5% level of significance that the manufacturing process is producing oversized ball bearings.	A handful of students identify this as a left tail test, resulting in a p-value that is more than 0.5. Note that for hypothesis testing, it does not make sense to have a p-value that is more than 0.5. As a rule of thumb, if $\bar{x} > \mu_0$, we should test $\mu > \mu_0$.

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9(iii)	For the new sample, $\bar{x} = 0.506$. However, $s^2 = \frac{1}{199} \left(2(0.08345) - \frac{(2(0.6))^2}{200} \right)$ $= 8.02513 \times 10^{-4}$ $Z_{calc} = 2.995$, p-value = 0.00137 Since p-value < 0.05, we will still reject H_0. The conclusion remains the same.	Combined sample is NOT pooled sample. Please do not use this formula at all! (It's for Further Math) For this part, candidates are supposed to either compute the p-value after the samples are combined, or give clearly supported reason why the p-value will be smaller.
10(i)	Let X denotes the diameter of bolt from manufacturer A. $X \sim N(1.56, 0.16^2)$ Let Y denotes the diameter of bolt from manufacturer B. $Y \sim N(\mu, 0.16^2)$ $P(Y < 1.52) = 0.242$ $P(Z < \frac{1.52 - \mu}{0.16}) = 0.242$ $\frac{1.52 - \mu}{0.16} = -0.6998836 \Rightarrow \mu = 1.63198 = 1.632$	Candidates are expected to show full working for this part as it is a 'show' question. Standardising is the preferred method. Candidates who used graphical method using did not explain the graphs used and how the final answer is attained. Tables should not be used when dealing with a non-integer value.
10(ii)	$W = X - Y \sim N(1.56 - 1.632, 0.16^2 + 0.16^2)$ $P(W < 0.1) = P(-0.1 < W < 0.1) = 0.326$	Quite a few candidates find the distribution of W instead of W, which is conceptually incorrect, and hence leading to a wrong answer.
10(iii)	$X_1 + X_2 + X_3 + X_4 + X_5 - 5Y \sim N(5(1.56) - 5(1.632), 5(0.16^2) + 5^2(0.16^2))$ $P(X_1 + X_2 + X_3 + X_4 + X_5 > 5Y)$ $= P(X_1 + X_2 + X_3 + X_4 + X_5 - 5Y > 0)$ $= 0.341$	Mistakes on calculating the correct variance were not as common this time round as compared to Midyear exams. However, poor representation of the variables is still commonly seen.
10(iv)	$P(X < 1.52) = 0.40129$ Prob. Req'd = $(0.44)(0.4012) + 0.56(0.242)$ $= 0.3120876$ $= 0.312$	Many candidates were unable to tackle this part. Again, as this is a 'show' question, candidates are expected to work out $P(X < 1.52) = 0.40129$.